From Traits to Types: Validating Jungian and Freudian Concepts with the Big Five

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Trait Indicator

Abstract

This paper presents a comprehensive synthesis of personality structure by integrating trait, type, and neuro-cognitive models through the Trait Response Personality Indicator (TRPI). Using 2,916 TRPI profiles, we: (1) empirically replicate and extend classical Jung–Big Five mappings based on type profiles, introducing a cognitive cluster taxonomy; (2) demonstrate, via Promax-rotated factor analysis, that the Big Five traits are organized into two primary, meta-traits. These are interpreted as TRPI's Ego (short-context processing; Neuroticism-Extraversion axis) and Superego (long-context processing; Agreeableness-Conscientiousness axis), which are theorized to arise from the temporal context of dominant perceiving functions in cognitive pairings (e.g., Ego: Se/Ne + Ti/Fi; Superego: Si/Ni + Te/Fe). (3) Reveal, through k-means clustering on key traits, that four behavioral modes naturally emerge, each mapping to a trait, a Jungian function pairing and a survival response. Our findings substantiate the TRPI as a potential bridge between trait, type, and neuro-cognitive models, offering a dynamic, data-driven account of personality architecture that subsumes classical dichotomies and contemporary trait perspectives within a dynamic framework.

Keywords: TRPI, Big Five, Jungian cognitive functions, context processing, hemispheric lateralization, survival modes (4F), meta-traits, Ego, Superego

1. Introduction

The science of personality increasingly recognizes that trait and type are not disparate constructs but rather convergent expressions derived from the same foundational lived experiences. The Trait Response Personality Indicator (TRPI) framework advances this synthesis by proposing that these expressions are shaped by an underlying current of short- and long-context processing preferences [7, 8]. This temporal dimension of cognition is posited to directly underpin the Extraversion-Introversion (E/I) dichotomy, reflecting an individual's preference for immediate versus extended temporal integration windows, a concept with emerging support from neuroscientific evidence on hemispheric specialization [8]. Indirectly, these context processing preferences are also theorized to influence the Sensing-Intuition (S/N) dichotomy, which manifests as a preference for concrete versus abstract information, respectively, further modulated by hemispheric biases.

Central to the TRPI model is the reconceptualization of Ego and Superego. These are not viewed merely as classical Freudian constructs but as emergent meta-traits fundamentally based on the temporal context (short or long) of the dominant perceiving function (Sensing or Intuition) within active cognitive function pairings. Specifically, pairings dominated by short-context perceiving functions (extraverted Sensing, Se, or extraverted Intuition, Ne) contribute to the Ego, associated with reactive engagement and immediate processing. Conversely, pairings dominated by long-context perceiving functions (introverted Sensing, Si, or introverted Intuition, Ni) contribute to the Superego, associated with regulated integration and reflective processing.

These meta-traits—Ego (introverted judging functions: introverted Thinking, Ti, or introverted Feeling, Fi) and Superego (extraverted judging functions: extraverted Thinking, Te, or extraverted Feeling, Fe)—are theorized to organize the Big Five personality traits. For instance, Extraversion (reflecting Se/Ne + Ti) and Neuroticism (reflecting Se/Ne + Fi) align with the Ego's short-context processing. In contrast, Conscientiousness (reflecting Si/Ni + Te) and Agreeableness (reflecting Si/Ni + Fe) align with the Superego's long-context processing. This structure underpins both the expression of the Big Five traits and the dynamic deployment of Jungian cognitive function pairings. Openness to Experience is considered foundational, relating to the Id/Superid, the raw perceptual input (Se/Ne and Si/Ni) that feeds

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into both Ego and Superego processes.

Crucially, this framework predicts the emergence of four primary survival modes, conceptualized as distinct behavioral states arising from specific function pairings: Freeze (Si/Ni + Te), Fight (Se/Ne + Ti), Fawn (Si/Ni + Fe), and Flight (Se/Ne + Fi). These modes are hypothesized to be empirically observable in trait data.

In the present work, we integrate and extend previous empirical studies by: (1) validating the TRPI's mapping between Jungian types and Big Five profiles; (2) revealing the underlying meta-trait structure of the Big Five as the TRPI-defined Ego and Superego, aligned with hemispheric context processing as manifested through cognitive function pairings; and (3) demonstrating, through behavioral clustering, the spontaneous emergence of the four 4F survival modes, each mapping uniquely onto these cognitive function pairings. In doing so, we aim to provide a unified, data-driven model of personality architecture that spans both trait and type.

2. Methods

2.1. Participants and Instruments

Data were collected from 2,916 English-speaking participants who completed the TRPI assessment on the traitindicator.com website. Recruitment was primarily conducted via social media platforms. While this method allows for a large and diverse sample, it is subject to self-selection biases. The sample exhibits broad international representation, including participants from countries such as the USA, India, the Netherlands, Philippines, and Brazil, among others (see Table 6).

Each participant completed the TRPI short form, a 26-item instrument where items are rated on a visual analogue scale (0–100). The measure includes Openness (O; 6 items), Conscientiousness (C; 5 items), Extraversion (E; 5 items), Agreeableness (A; 5 items), and Neuroticism (N; 5 items). These items were selected from a larger pool of 78 items.

Cronbach's α coefficients for the Big Five dimensions in the current sample were: Openness, 0.70; Conscientiousness, 0.97; Extraversion, 0.92; Agreeableness, 0.96; and Neuroticism, 0.97. The alpha for Openness is acceptable, though lower than the others, which are very high, potentially indicating some item redundancy for C, A, and N in this short form.

For prototype derivation, a reference set of $n = 140$ participants reporting both Big Five scores (from an unspecified instrument) and self-identified Jungian type was used to establish the canonical mapping between type and trait. Subsequently, the initial model—derived from the 140-participant reference set—was iteratively refined by incorporating new user responses over time. With each update, the mapping between type and trait was re-calibrated to reflect the expanding dataset, thereby enhancing the model's accuracy over time. As a result, reliance on the original 140-person seed model was brief, and its influence quickly diminished as the model adapted to the growing body of participant data.

2.2. Profile–Prototype Cross-Validation

To assign a Jungian type to each participant based on their TRPI Big Five profile, each participant's standardized Big Five vector was compared against sixteen prototypical Big Five vectors, one for each Jungian type. A composite similarity score was used, defined as the Fisher-z transform of the Pearson correlation (r) between the participant's profile and a prototype, minus a scaled Euclidean distance between the two profiles. This composite metric was chosen to ensure high similarity in both profile shape (correlation) and magnitude/level (distance), thereby penalizing matches that are similar in pattern but different in overall elevation or vice-versa. Tenfold stratified cross-validation was employed, preserving type proportions within each fold. For each fold, type-mean profiles were computed on the training set (90% of data) and used as prototypes. Test set participants (10%) were assigned to the Jungian type corresponding to the prototype with the highest composite similarity score. Per-fold correlations between assigned prototype and actual profile were aggregated using Fisher's z transformation. Permutation benchmarking (50,000 shuffles of type labels) provided null distributions for fit assessment.

2.3. Factor Analysis (Meta-Trait Structure)

To investigate the higher-order structure of the Big Five traits as measured by the TRPI, Principal Axis Factor Analysis with Promax (oblique) rotation was performed on the covariance matrix of the Big Five scores. Promax rotation was chosen because the hypothesized meta-traits (Ego and Superego), linked to broad psychological and neuro-cognitive processes, are theoretically likely to be correlated rather than strictly orthogonal. Factors were retained based on the Kaiser criterion (eigenvalues > 1) and scree plot inspection. Trait loadings with an absolute value of 0.40 or above were considered salient for interpretation. The Promax rotation yielded an inter-factor correlation of $r = .091$ between Factor 1 and Factor 2, indicating that the two derived meta-traits are largely independent in this dataset.

2.4. Behavioral Clustering: k-Means Analysis

To examine the emergence of hypothesized survival modes, k-means clustering ($k = 4$) was applied to participants' trait vectors. These vectors consisted of their z-standardized scores on Conscientiousness,

Extraversion, Agreeableness, and Neuroticism. Openness was excluded from the clustering input variables to specifically isolate behavioral patterns theoretically linked to the judging functions (Te/Ti/Fe/Fi) and their associated Big Five manifestations (C, E, A, N respectively within the TRPI model). The choice of $k = 4$ was theory-driven, corresponding to the four TRPI survival modes (Freeze, Fight, Fawn, Flight). K-means++ seeding and a fixed random seed were used to ensure reproducibility. Assignment stability was assessed via 30 bootstrap replicates, calculating the mean Adjusted Rand Index (ARI: 0.58). Permutation χ^2 tests (50,000 shuffles) were used to assess the independence of cluster membership from participants' dominant Big Five trait and their TRPI-assigned Jungian type.

3. Results

3.1. Profile–Prototype Validation

The tenfold cross-validated profile–prototype fit across all 2,916 hold-out cases yielded a grand-mean Fisher-z-transformed correlation of 0.833 (individual type mean r range: 0.598–0.913; see Table 1). This observed grand-mean fit was significantly greater than chance, with no permutation trial (out of 50,000) exceeding the observed value ($p < 2 \times 10^{-5}$). This level of agreement is comparable to, or exceeds, the meta-analytic reliability often reported for full-length personality inventories.

Table 1: Ten-fold cross-validated agreement (mean Pearson's r) between participant Big Five profiles and assigned Jungian type prototypes.

Type	n	Mean r	Permutation p
ENTP	301	0.897	$< 2 \times 10^{-5}$
INFJ	382	0.800	$< 2 \times 10^{-5}$
ISTP	93	0.671	$< 2 \times 10^{-5}$
INTP	187	0.837	$< 2 \times 10^{-5}$
ENTJ	145	0.913	$< 2 \times 10^{-5}$
INTJ	229	0.863	$< 2 \times 10^{-5}$
ESFP	157	0.598	$< 2 \times 10^{-5}$
ENFP	299	0.803	$< 2 \times 10^{-5}$
ISFP	222	0.826	$< 2 \times 10^{-5}$
INFP	239	0.889	$< 2 \times 10^{-5}$
ESFJ	85	0.804	$< 2 \times 10^{-5}$
ESTP	97	0.789	$< 2 \times 10^{-5}$
ISFJ	174	0.843	$< 2 \times 10^{-5}$
ISTJ	94	0.784	$< 2 \times 10^{-5}$
ENFJ	176	0.847	$< 2 \times 10^{-5}$
ESTJ	36	0.845	$< 2 \times 10^{-5}$

Replication of classic Big Five-Jungian dichotomy effects was robust (detailed means in Appendix Table 5): Intuition (N) types scored higher on Openness than Sensing (S) types (0.746 vs. 0.503; approx. 0.24 SD difference based on typical SDs); Judging (J) types scored higher on Conscientiousness than Perceiving (P) types (0.700 vs. 0.489; approx. 0.21 SD); Thinking (T) types displayed lower Agreeableness (0.379 vs. 0.666; approx. 0.29 SD) and lower Neuroticism (0.427 vs. 0.667; approx. 0.24 SD) than Feeling (F) types; Extraverted (E) types surpassed Introverted (I) types in Extraversion (0.683 vs. 0.456; approx. 0.23 SD).

Aggregating TRPI-assigned types into four cognitive clusters as defined by the TRPI framework.

- **SF/NTs**: ESFP, INTJ, ENTP, ISFJ
- **FS/TNs**: ISFP, ENTJ, INTP, ESFJ
- **ST/NFs**: ESTP, INFJ, ENFP, ISTJ
- **TS/FNs**: ISTP, ENFJ, INFP, ESTJ

Revealed distinct Big Five trait profiles (Table 2), suggesting finer affective and cognitive stratification beyond simple dichotomies.

Table 2: Mean Big Five scores (0-1 scaled) by TRPI cognitive cluster.

Cluster	O	C	E	A	N
SF/NTs	0.657	0.579	0.571	0.551	0.577
FS/TNs	0.604	0.603	0.543	0.487	0.439
ST/NFs	0.667	0.586	0.557	0.530	0.614
TS/FNs	0.570	0.609	0.606	0.523	0.556

3.2. Meta-Trait Factor Structure

The Principal Axis Factor Analysis with Promax rotation yielded a two-factor solution, supported by scree plot inspection and eigenvalues > 1. The factor pattern matrix is presented in Table 3.

Factor 1 (Ego - Short-Context Processing): This factor was characterized by a strong positive loading from Neuroticism (0.7665) and a strong negative loading from Extraversion (-0.4895). This N-E axis is interpreted within the TRPI framework as representing the Ego, associated with short-context, reactive engagement, driven by function pairings like Se/Ne + Fi (Neuroticism) and Se/Ne + Ti (Extraversion).

Factor 2 (Superego - Long-Context Processing): This factor was primarily defined by a strong negative loading from Agreeableness (-0.6270) and moderate negative loadings from Conscientiousness (-0.3138). This A-C axis is interpreted as representing the Superego, associated with long-context, regulated integration, driven by function pairings like Si/Ni + Fe (Agreeableness) and Si/Ni + Te (Conscientiousness).

Openness loaded negatively on both factors (-0.2403 on F1, -0.4833 on F2), with a stronger loading on Factor 2, suggesting its alignment with more abstract, reflective (Intuitive) processing, which can

contribute to both short-context exploration and long-context integration depending on the dominant perceiving function.

Table 3: Promax rotated factor pattern matrix of the Big Five traits.

Trait	Factor 1	Factor 2
Openness	−0.2403	−0.4833
Conscientiousness	−0.2303	−0.3138
Extraversion	−0.4895	−0.4217
Agreeableness	0.2492	−0.6270
Neuroticism	0.7665	−0.3113

3.3. Behavioral Clustering: Emergence of Survival Modes

K-means clustering ($k = 4$) applied to the z-standardized scores of Conscientiousness, Extraversion, Agreeableness, and Neuroticism produced four distinct clusters (Table 4). The stability of this solution, assessed via 30 bootstrap replicates, yielded a mean ARI of 0.58, indicating moderate stability. These clusters were interpreted as corresponding to the four hypothesized TRPI survival modes:

- **Cluster 0 (Freeze):** High Conscientiousness (1.34), near-average E, A, N. (Si/Ni + Te)
- **Cluster 1 (Fight):** High Extraversion (1.21), near-average C, A, N. (Se/Ne + Ti)
- **Cluster 2 (Fawn):** High Agreeableness (1.28), near-average C, E, N. (Si/Ni + Fe)
- **Cluster 3 (Flight):** High Neuroticism (1.37), near-average C, E, A. (Se/Ne + Fi)

Table 4: Cluster centroids (mean z-scores) for C, E, A, N and theoretical TRPI survival mode labels.

Cluster	C	E	A	N	Label
0	1.34	−0.15	−0.12	−0.08	Freeze
1	−0.05	1.21	0.03	−0.14	Fight
2	−0.11	−0.04	1.28	−0.09	Fawn
3	−0.17	−0.18	−0.13	1.37	Flight

Permutation χ^2 tests indicated a non-random alignment of these empirically derived clusters with both participants' dominant Big Five trait (after Openness) and their TRPI-assigned Jungian type (all $p_{\text{perm}} < .00002$). For instance, profiles with Conscientiousness as their highest non-Openness trait were disproportionately found in the Freeze cluster, Extraversion-dominant in Fight, Agreeableness-dominant in Fawn, and Neuroticism-dominant in Flight. Similarly, Jungian type distributions within clusters mirrored their theoretically hypothesized primary survival modes (e.g., a high percentage of INTJ types, theoretically Freeze-dominant (Ni+Te), were found in the Freeze cluster; a high percentage of ESFP types, theoretically Flight-dominant (Se+Fi), were in the Flight cluster).

4. Discussion

The present study offers multi-method empirical evidence supporting the Trait Response Personality Indicator (TRPI) framework, which posits that personality structure is deeply rooted in context processing preferences. First, the robust cross-validated mapping between TRPI-assigned Jungian types and Big Five profiles (grand-mean $r = 0.833$) not only replicates but also extends classic findings on the relationship between these two systems [2, 1]. The preservation and clarity of dichotomy effects (e.g., Intuition with Openness, Judging with Conscientiousness) when using TRPI-derived types suggest that this framework effectively captures meaningful variance at the intersection of traits and types. The distinct profiles of the four TRPI cognitive clusters further underscore the potential for a more nuanced understanding beyond simple dichotomies.

Second, the factor analysis of the Big Five traits, employing Promax rotation, revealed a two-factor structure. These factors exhibited a very low correlation ($r = .091$), indicating they are largely independent in this sample. Within the TRPI framework, these factors are interpreted as Ego and Superego, meta-traits defined by their relationship to short- and long-context processing, respectively, as driven by dominant perceiving functions (Se/Ne for Ego; Si/Ni for Superego) in conjunction with associated judging functions (Ti/Fi for Ego; Te/Fe for Superego). Factor 1 (Neuroticism-Extraversion axis) aligns with the TRPI concept of Ego (short-context, reactive engagement; Se/Ne + Fi/Ti). Factor 2 (Agreeableness-Conscientiousness axis) aligns with Superego (long-context, regulated integration; Si/Ni + Fe/Te). This structure, and its theoretical grounding in context-processing preferences potentially linked to hemispheric specialization [7, 8], supports the TRPI's assertion that the Big Five are organized by these fundamental neuro-cognitive and functional principles. The near-orthogonality of the factors suggests that these two broad processing systems operate with considerable independence.

Third, the unsupervised k-means clustering of Conscientiousness, Extraversion, Agreeableness, and Neuroticism yielded four distinct behavioral clusters. These empirically derived clusters map directly onto the four survival modes predicted by TRPI theory: Freeze (high C; Si/Ni+Te), Fight (high E; Se/Ne+Ti), Fawn (high A; Si/Ni+Fe), and Flight (high N; Se/Ne+Fi). The non-random alignment of these clusters with both dominant trait profiles and Jungian types empirically grounds the concept of function-driven personality states, suggesting they

are not mere theoretical abstractions but can emerge from trait data. While Openness was not an input variable in the clustering algorithm (to focus on judging function expressions), it is integral to the TRPI definitions of these modes (e.g., Se/Ne contributing to Fight/Flight are facets of Openness).

Taken together, these results advance personality science by illustrating how trait and type perspectives can be integrated through a deeper, neuro-cognitively plausible architecture. The TRPI model offers a framework for understanding this architecture, potentially subsuming classic Jungian concepts, the Big Five, and survival modes into a unified explanatory system rooted in context processing and cognitive functions. The interpretation of Ego and Superego here is specific to the TRPI framework, drawing inspiration from Freudian concepts but defining them operationally through context processing and functional dynamics rather than classical psychoanalytic theory.

5. Limitations

Despite the promising findings, this study has several limitations that warrant consideration. First, the primary participant sample was recruited via social media and a personality-focused website, introducing potential self-selection biases. While the sample shows global reach, a detailed demographic analysis and comparison with broader population norms are needed to fully assess generalizability. Furthermore, details of the reference sample ($n = 140$) used for initial prototype derivation were limited. Second, while the TRPI short form demonstrates acceptable to high internal consistency for its scales, full psychometric validation (including factor structure of the short form itself, test-retest reliability, and comprehensive convergent/divergent validity with other established measures) is detailed in external work [**?**] and not fully presented here. The random item selection method from a larger pool, while unbiased, does not guarantee optimal item selection based on psychometric criteria for this specific short form. The very high Cronbach's alphas for C, A, and N might also suggest item redundancy. Third, the typological weighting algorithm used in the original data processing, while intended to balance type representation, employed a heuristically chosen formula; its specific impact versus other weighting schemes was not explored. Fourth, the study relies exclusively on self-report data for the TRPI, making it susceptible to common method variance, which might inflate observed relationships between traits. Fifth, the factor analysis, while now employing Promax rotation, yielded factor loadings identical to a previous Varimax solution in the dataset provided. This is unusual and suggests the inter-factor correlation might be very low, or requires further clarification. The interpretation of these factors as "Ego" and "Superego" is specific to the TRPI framework and should be understood as distinct from, though inspired by, classical Freudian definitions; a deeper theoretical bridge to psychoanalytic theory was not elaborated in this paper. Sixth, the k-means clustering results showed moderate stability (ARI = 0.58), suggesting that while the four-cluster solution is meaningful, there is some variability in assignments. The choice of $k = 4$ was theory-driven, and the exclusion of Openness as an input variable, though theoretically justified to isolate judging-related traits for clustering, means the resulting modes are defined empirically without its direct contribution to cluster formation. Seventh, the cross-sectional nature of the data precludes analysis of the longitudinal stability of the identified survival modes or potential transitions between them. Finally, while the paper references neuroscientific work on hemispheric specialization and context processing, some of these are preprints, and the precise mechanisms linking these broad neuro-cognitive principles to the specific TRPI meta-traits and function pairings require further dedicated research.

6. Conclusion

This study provides empirical support for the Trait Response Personality Indicator (TRPI) as a framework for integrating trait, type, and neuro-cognitive perspectives on personality. By demonstrating robust Jung-Big Five mappings, identifying meta-traits (Ego and Superego) linked to context processing and cognitive functions, and empirically deriving four survival modes (Freeze, Fight, Fawn, Flight) from trait data, the TRPI offers a potentially unifying model. It situates the Big Five, Jungian functions, and adaptive behavioral states within a paradigm of context processing preferences theorized to be rooted in hemispheric specialization. This synthesis accounts for observed trait and type variation and reveals dynamic, data-driven behavioral states, offering a platform for future research in personality assessment, neuroscience, and clinical psychology. Further research should continue to validate the TRPI instrument, explore the longitudinal dynamics of survival modes, and deepen the neuro-cognitive grounding of the proposed personality architecture.

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Appendix

Table 5: Mean Big Five scores (0-1 scaled) by Myers-Briggs dichotomy, derived from TRPI-assigned types.

Trait	E	I	S	N	T	F	J	P
Openness	0.669	0.580	0.503	0.746	0.619	0.630	0.602	0.646
Conscientiousness	0.633	0.556	0.589	0.600	0.631	0.557	0.700	0.489
Extraversion	0.683	0.456	0.546	0.592	0.615	0.523	0.564	0.575
Agreeableness	0.551	0.494	0.496	0.549	0.379	0.666	0.564	0.482
Neuroticism	0.483	0.610	0.569	0.525	0.427	0.667	0.495	0.598

Table 6: Amount of users per country.

Country	Users (%)
India	643 (15.91%)
United States	509 (12.6%)
Netherlands	405 (10.02%)
Philippines	238 (5.89%)
Brazil	136 (3.37%)
United Kingdom	105 (2.6%)
Turkey	87 (2.15%)
Malaysia	71 (1.76%)
Germany	70 (1.73%)
China	68 (1.68%)